PERMEABLE INFRASTRUCTURE FOR PEDESTRIAN ROUTES

Installation Guide

1. Scope of Application

Flexus AP-Series is intended for architectural-grade permeable pavement on pedestrian routes, with a controlled aggregate palette and a stone-forward surface finish, including:

- Botanical gardens
- Zoo pathways
- Courtyards and campus circulation
- Nature preserves

Flexus AP-Series is not designed for vehicular traffic, emergency vehicle loading, or structural pavement applications unless specifically engineered.

2. System Overview

Flexus is a poured-in-place permeable surface composed of:

- Washed specialty aggregate
- Colored recycled tire rubber granules
- Single-component moisture-cure polyurethane binder

Installed over a properly prepared aggregate base.

Standard thicknesses:

MODEL	INSTALLED THICKNESS
AP-150	1.5 in (38 mm)
AP-200	2.0 in (51 mm)
AP-250	2.5 in (64 mm)

The system is designed to allow vertical infiltration through the surface and base layer when installed over properly prepared subgrade. An infiltration rate is not published. ASTM C1701 testing is in progress. See section 18.

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3. Typical Section

A typical installation includes:

1. Flexus surface layer
2. Compacted clean aggregate base
3. Optional geotextile separator
4. Compacted subgrade

Base thickness should be determined by site conditions and soil classification. Flexus pavement does not correct unstable or saturated subgrade conditions.

4. Subgrade Preparation

Subgrade shall be:

- Stable and well draining
- Free of organic material
- Free of standing water
- Compacted to a minimum of 95 percent Standard Proctor (ASTM D698)

Areas exhibiting pumping, soft soils, or high moisture content must be remediated prior to installation. Recommended surface slope: 1 to 2 percent for positive drainage where required. Installation over frozen or saturated subgrade is not permitted.

5. Aggregate Base Requirements

Base aggregate shall be clean, washed, angular stone; free of fines and clay; typically ASTM No. 57 or similar gradation. Minimum recommended base thickness for pedestrian applications: 4 in to 6 in compacted. Base shall be compacted in controlled lifts. Crusher run or materials containing fines are not permitted. Base shall provide uniform support and positive drainage.

Reference details show a typical 4 in ASTM No. 57 base and an optional 6 in ASTM No. 2 subbase. The above-grade detail shows edge restraint at a minimum 8 in depth. The engineer of record sets the actual section.

6. Edge Condition Requirements

A. Rigid edge restraint (preferred where the surface is exposed above adjacent grade). Rigid edge restraint is required where the surface is exposed above adjacent grade. Acceptable systems include steel landscape edging, aluminum restraint systems, cast-in-place concrete curb, and engineered composite restraint systems. Edging must be securely anchored prior to surface placement. Reference details show edge restraint at a minimum 8 in depth, installed flush with the finished surface, with passive relief drainage where applicable.

B. Subgrade or backfilled beveled edge. Where the Flexus surface terminates below adjacent grade and is fully laterally confined by compacted soil or aggregate, rigid edge restraint is not required. The edge shall be a 45-degree bevel, compacted, with adjacent soil properly compacted, and lateral confinement maintained. Exposed, unsupported edges are not permitted.

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7. Environmental Conditions

Installation shall proceed only under the following conditions:

- Ambient and substrate temperature: 50°F (10°C) and rising, and not more than 104°F (40°C)
- Dry base and subgrade. Do not apply over wet or damp substrates
- No precipitation during installation
- No freezing temperatures within 24 hours

Elevated humidity or substrate moisture may adversely affect curing.

8. Mixing Requirements

Flexus mix design must follow approved ratios. Typical AP-200 (2.0 in) example:

- Solids per square foot: approximately 10.34 lb
- Binder content: 11 to 12.5 percent by weight of total solids (minimum 11 percent by weight)
- Aggregate to recycled tire rubber: 50:50 by volume

Materials must be mechanically mixed until homogeneous. Unauthorized alteration of mix ratios voids system warranty. Solids per square foot for the 1.5 in and 2.5 in models are not published here; confirm quantities through specification support.

9. Placement and Working Time

Mixed material shall be placed immediately after batching. Maximum time from completion of mixing to placement: 20 minutes at 70°F (21°C). Working time is reduced at elevated temperatures. Do not re-temper or add binder after initial mixing. Material shall not be placed if consistency indicates premature curing. Cold joints must be minimized. Where unavoidable, edges shall be cut clean prior to resuming installation.

10. Execution Sequence

Installation should proceed continuously to minimize cold joints.

1. Verify subgrade and base compaction (subgrade: minimum 95 percent Standard Proctor, ASTM D698).
2. Confirm environmental conditions in section 7.
3. Confirm edge restraint or beveled confinement is in place and anchored as required.
4. Stage materials and confirm quantities against the approved mix design.
5. Mechanically mix aggregate, recycled tire rubber, and binder until homogeneous.
6. Place material within edge restraints immediately, within the working time in section 9.
7. Screed to the specified model thickness (1.5 in, 2.0 in, or 2.5 in).
8. Finish the surface uniformly. Surface finish shall be uniform and consistent with the approved architectural sample and aggregate blend.
9. Protect the surface during curing (section 11).

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11. Curing

Typical cure schedule (temperature dependent):

- Initial set: 4 to 8 hours
- Light pedestrian traffic: 24 hours minimum
- Full cure: 48 to 72 hours

Surface must be protected from moisture during initial cure.

12. Submittals and Mockup

Before placement, submit the following for the specified series and model, or confirm they have been provided:

- Product overview (cut sheet) and this installation guide
- PDF reference details (illustrative; not for construction without a registered engineer's signature)
- A cured sample of the specified palette or blend
- Approved mix design (ratios in section 8)
- Installer identification (Flexus Materials Ltd Co, or a contractor supplied with the Flexus system and this install guidance)

A CSI 32 12 43 (Porous Flexible Paving) three-part section and a submittal checklist are available on request through specification support. They are not a substitute for the project specification or the engineer of record.

Install a mockup of the specified model and palette for approval before full production. Surface finish shall reflect the approved sample and aggregate blend. Mockup size is set with the owner or in the project specification.

13. Field Quality Control

Checks that are part of this guide:

- Subgrade compaction: minimum 95 percent Standard Proctor (ASTM D698)
- Environmental conditions: section 7
- Working time and mix ratios: sections 8 and 9
- Installed thickness: match the specified model (1.5 in, 2.0 in, or 2.5 in). A numeric thickness tolerance is not published; verify thickness against the specified lift as the work proceeds
- Finish: uniform and consistent with the approved sample

If the project specification requires in-place infiltration testing, use ASTM C1701. Flexus has not published an acceptance rate; the engineer of record sets project criteria. System testing under ASTM C1701 is in progress (section 18).

14. Accessibility and Slope

When properly installed, Flexus may support ADA slope conditions. Recommended maximum cross slope: 2 percent. Longitudinal slope per ADA standards. Final compliance is the responsibility of the project designer.

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15. Maintenance

Routine maintenance includes periodic debris removal, leaf blower cleaning, and edge restraint inspection. Annual inspection is recommended. Avoid metal snow plows or unapproved chemical deicers. In planted settings, keep leaf litter and soil from building up on the surface so drainage paths stay open. A published restoration method and a list of approved deicers are not included in this guide; contact specification support for project advice.

16. Repair

Localized repairs may be completed by removing the affected area, cleaning the base, installing new material, and blending perimeter edges. Minor color variation may occur at repair interfaces.

17. Limitations

> Flexus is not structural pavement and is not intended for vehicular loading. Performance depends on proper subgrade preparation, drainage, and installation practices. Installer qualification and adherence to approved procedures are required.

18. Data Available and Data in Testing

Use this table when writing a specification. Status is current as of November 2026.

PROPERTY	METHOD	STATUS
Surface infiltration	ASTM C1701	In testing
Slip resistance	ASTM E303	In testing
Tensile strength and elongation of the cured surface	ASTM F1551 / ASTM D412	In testing
UV exposure	ASTM G154	In testing
Abrasion resistance	ASTM C501	In testing
Flammability (methenamine pill)	ASTM D2859	In testing
Leachable metals, AP-Series, RCRA 8 metals	EPA 1311 TCLP and EPA 1312 SPLP	Available on request (full report only)

> No result is published until a Flexus-named report is complete. Do not infer a rate, rating, or pass/fail from this table.

19. Disclaimer

> This document provides general installation guidance. Site conditions vary. Final engineering design, base thickness determination, and compliance verification are the responsibility of the project engineer and installer.

For technical assistance or project-specific review, contact Flexus Materials Ltd Co, support@flexusco.com, www.flexusco.com.

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